

The Model Sees the Market. We Tried to Find Out Why It Trades the Way It Does.

A nine-phase investigation into whether a trading model's internal market picture actually drives its decisions --- and what happened when we kept pushing until the evidence gave us an honest answer.

DATE	SPAN	RESULT	STATUS
6 April 2026	Nine phases	Real internal market picture; no single decisive cause	Validated through path test

TOP LINE

We found a real internal market picture inside the model, and we showed that context can change that picture. But the two strongest candidate signals did not hold up as single clean causes of the final trading choice.

EARLY CONTEXT EFFECT

+6.1 pp

Risk framing changed the market read when it appeared before the market block.

OPPORTUNITY EFFECT

+7 pp

Opportunity framing produced a similarly sized shift.

STRONGEST ROBUSTNESS RESULT

12 / 12

Targeted edits beat matched random edits in every main comparison.

FINAL SOURCE-MATCH TEST

+4.17 pp

The stronger signal helped a little, but not enough to count as the cause of the final choice.

Why This Study Matters

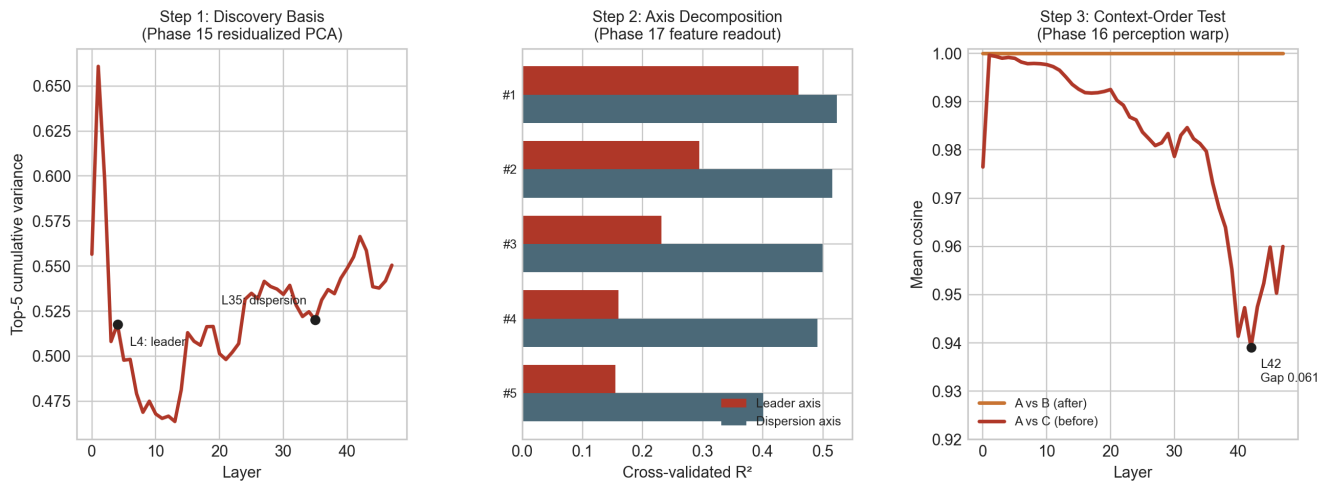
Most interpretability work stops at discovery. You find a pattern inside a model, you name it, and the pattern becomes the explanation.

That is not enough. A signal that looks meaningful might just be wallpaper — something the model computes along the way but never actually leans on when it reaches for a decision. Finding a readable internal pattern and proving that it drives behavior are very different things. Confusing them is the easiest mistake in this field, and the most consequential.

This study applied a sequence of increasingly hard tests to the same set of internal signals:

- find a pattern
- confirm it survives after removing prompt-format confounds
- test whether earlier context bends it
- test whether editing it changes behavior
- test whether restoring it moves behavior back toward a known source
- test whether a surgical lesion-and-rescue reveals a causal path

That last test is where many appealing stories fall apart. In our case, that is largely what happened — and that is why the result is useful.



The methodology at a glance. Each phase tightened controls and raised the standard of evidence required before the next test.

Phase Summary

Phase	What we asked	What we tested	What held up	Confidence
1	Discovery	Does the model show a clean internal market read?	Two promising patterns appeared: one linked to a standout asset and one to market unevenness. Some signal was still mixed with prompt-format effects.	promising, not yet clean
2	Format De-contamination	Do the same patterns survive after removing prompt-shape effects?	The average market read became much cleaner. Both main patterns survived. The end-of-market read was still noisier.	strong for the average market read
3	Context Sensitivity	Does earlier context change how the model reads the same market?	Context before the market changed the model's read. Context after did not.	strong
4	Axis Decomposition	What do the recurring internal patterns mean in plain market terms?	One pattern looked like a standout asset with strong activity. The other looked like the market's overall unevenness.	good description, not a full explanation

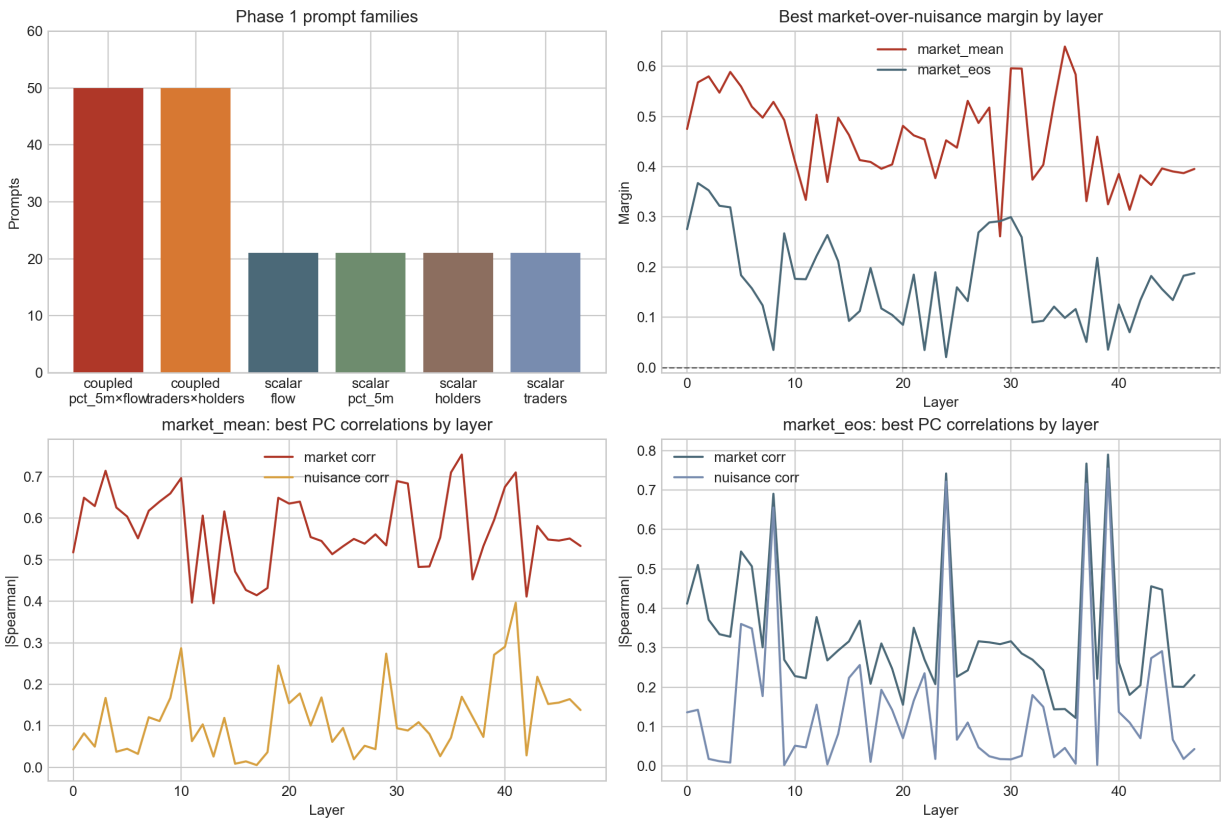
Phase	What we asked	What we tested	What held up	Confidence
5	Intervention	If we weaken one of these signals, does the model behave differently?	Single directions were weak. Broader groups mattered more. Pointed away from a one-switch story.	suggestive, not decisive
6	Controlled Intervention	With tighter controls, does the signal hold?	Targeted edits mattered slightly more than random edits, but the gap was modest.	real signal, still exploratory
7	Robustness Battery	Are targeted edits actually special, or would any perturbation of the same size produce the same effect?	In all 12 main comparisons, targeted edits were less disruptive than matched random edits. The signal was real and selective.	strong for selectivity, not yet causal
8	Restoration	If we put matching signal back in, does behavior move back toward the source example?	The stronger candidate helped a little. The weaker did not. Neither looked like a decisive cause of the final choice.	enough to rule out a strong causal claim
9	Path Validation	If we damage the early signal and rescue it downstream, does behavior recover?	Interventions applied cleanly, but the behavior surface was too sparse to measure action-choice changes with confidence.	mechanically sound, behaviorally inconclusive

Notice the confidence column: early optimism, growing precision, and a final reckoning with what the evidence actually supports. The rest of this report fills in the data behind each row.

Discovery: The Model Builds An Internal Market Picture

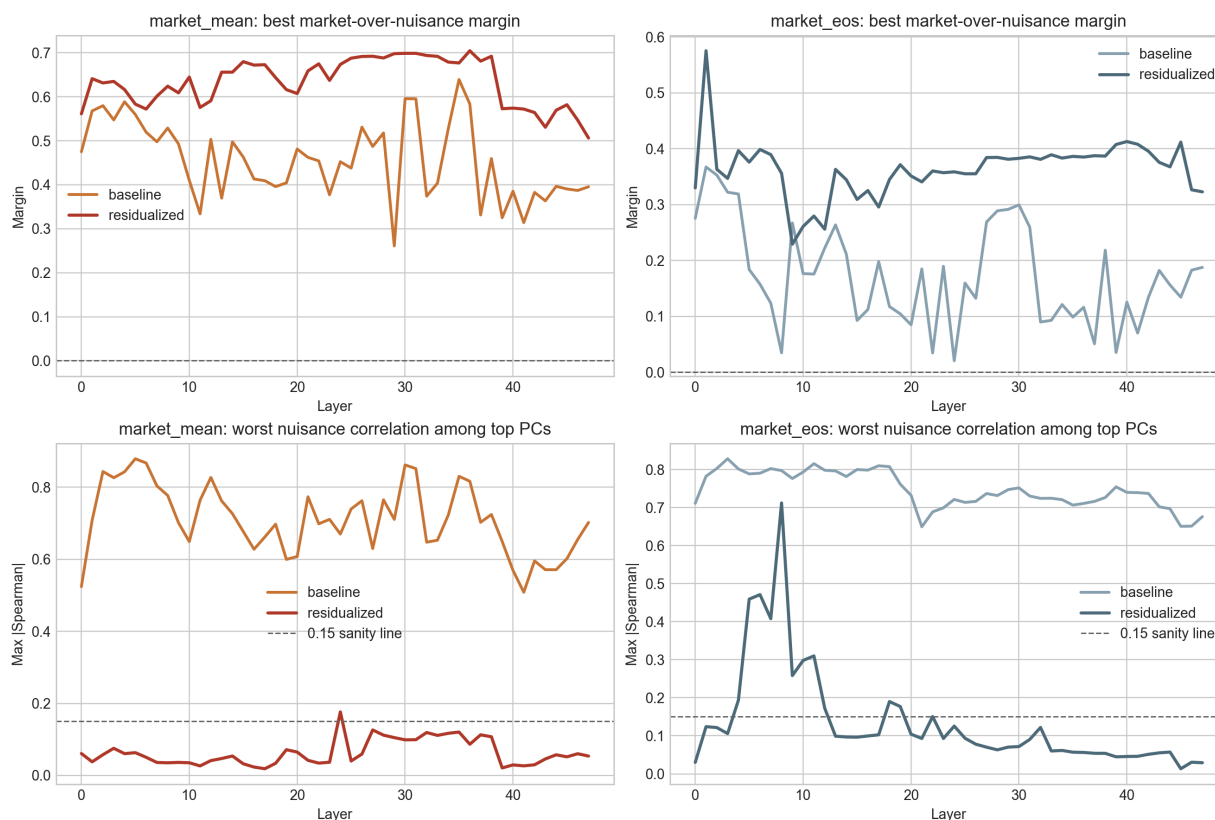
The synthetic prompts mirror the real task: the same system prompt style, the same section order, the same market framing, six assets per market. The question was whether the model's hidden activations contain a recognizable picture of the market it just read.

They do. Two strong candidate patterns appeared consistently across the prompt set.



The first discovery pass found clear market-linked internal patterns. Two recurring signals stood out, both linked to visible market features.

Some of that initial signal was tangled with prompt-format effects. After removing the part of the internal signal predictable from prompt shape, the same two patterns survived — cleaner and stronger. The average-market read was especially robust. The end-of-market read was noisier.



After removing format confounds, the model's average-market summary held up cleanly. The same two candidate patterns survived the decontamination.

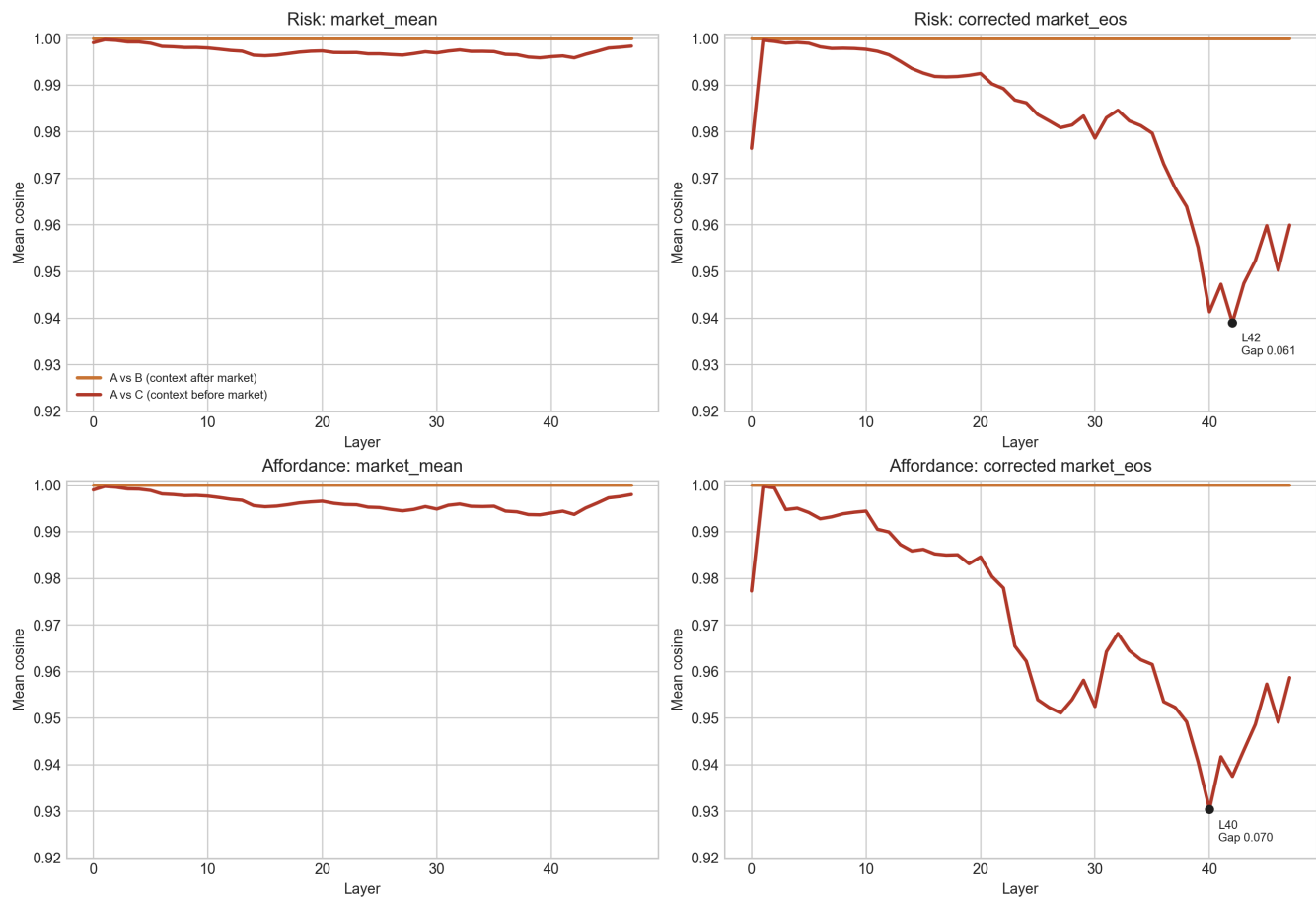
That gave us two confirmed internal signals worth testing further.

Context Sensitivity: Prior Framing Changes The Market Read

Is the model's market picture fixed, or does it depend on what the model has already been told?

We held the market constant and moved the same block of additional context either before or after the market section.

When the context came after the market, the market read stayed the same. When the same context came before the market, the model's reading shifted.



The same context matters when it appears before the market, not when it appears after. This is direct evidence that the model's first impression of the market depends on prior framing.

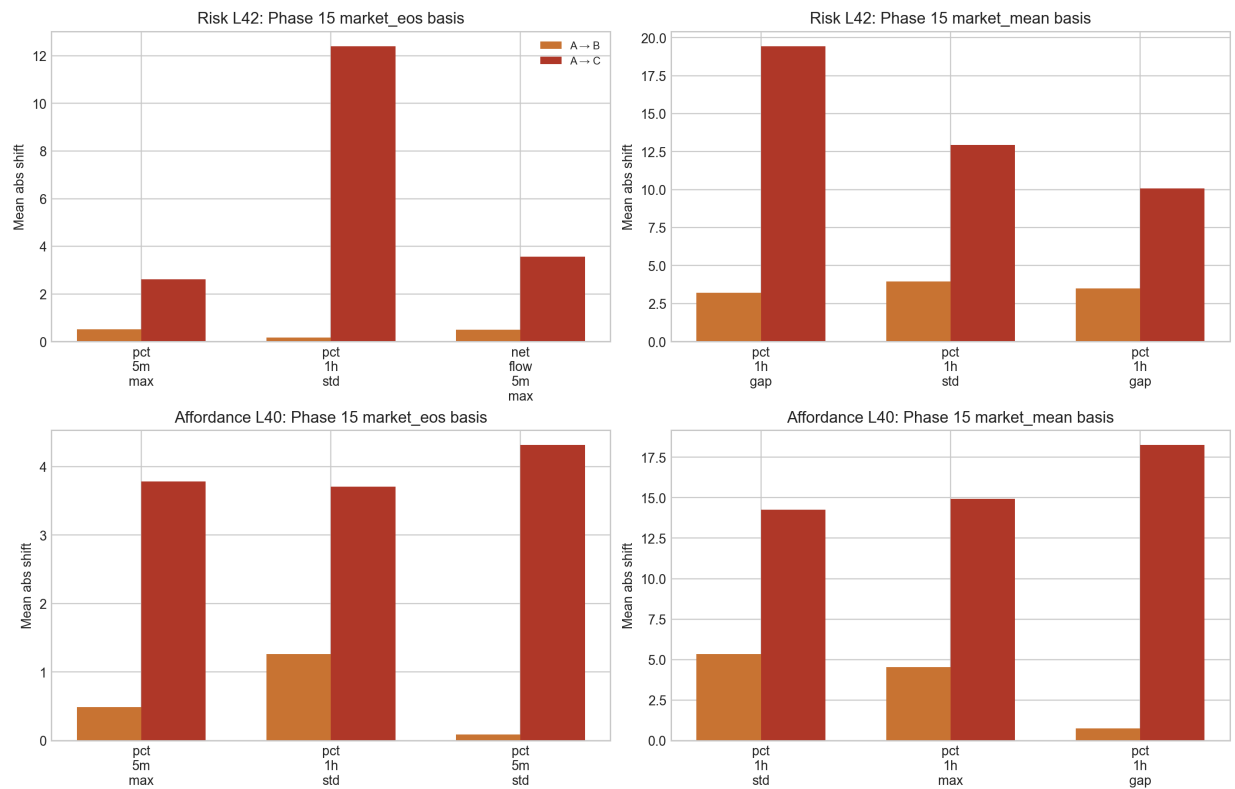
The size of that shift was not enormous, but it was real:

- risk framing produced a gap of **+6.1 pp**
- opportunity framing produced a gap of **+7 pp**

The key result is the asymmetry:

- after-market context did not retroactively rewrite the market picture
- before-market context changed how the model finished reading the market block

The model's market read is not a passive snapshot. It is shaped by the lens through which the model approaches the data. Change the lens, and the same numbers land differently.



The internal basis shifted measurably when context appeared before the market. When the same context appeared after, the basis stayed put. This is the fingerprint of a framing effect, not a retroactive rewrite.

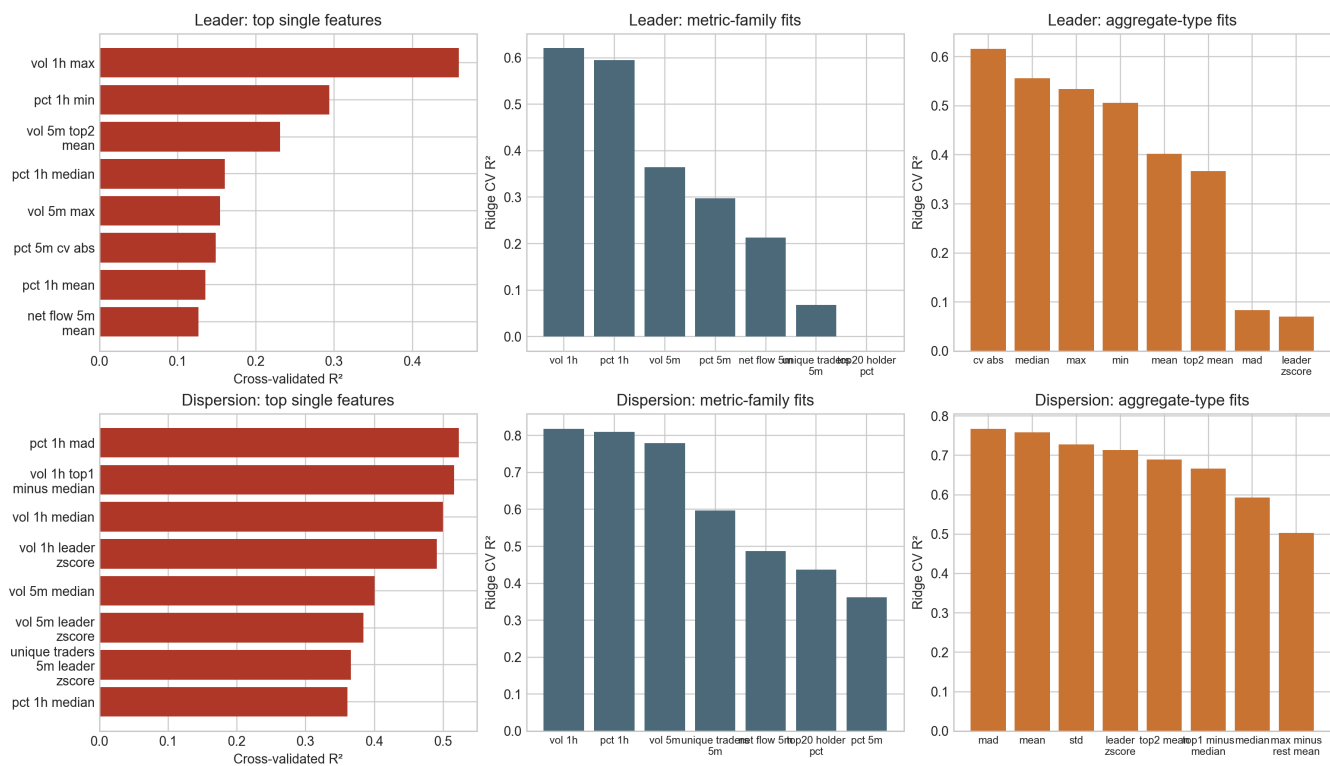
Axis Decomposition: What The Two Signals Mean

What do the two strongest recurring internal patterns correspond to in the visible market?

To answer this, we matched the model's internal patterns only against features a human reader could also see in the prompt — no hidden labels, no synthetic ground truth.

The result was intuitive, but not reductive:

- one recurring pattern looked like a standout asset with strong activity around it
- the other looked like how uneven the market was overall



The two strongest internal patterns line up with ordinary market descriptions, but neither collapses into one perfect formula. That turned out to matter later.

Two concrete markers stood out:

- the clearest visible clue for the first pattern was **vol_1h_max**
- the clearest visible clue for the second pattern was **pct_1h_mad**

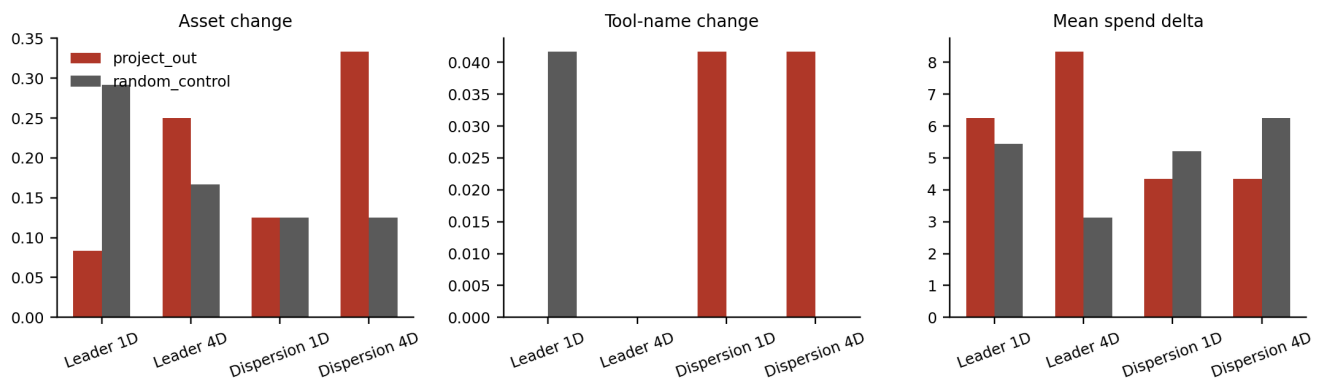
These visible features are not the full explanation — but they show that the model's hidden market summary can be described in ordinary market language.

The natural next question: do these signals actually drive the model's decisions, or are they just a readable byproduct?

Intervention: Does Editing The Signal Change The Choice?

We weakened the candidate signals directly inside the model's hidden state and measured whether the trading choice changed.

Single-direction edits were weak. Broader edits across a cluster of related directions produced stronger effects.



Single narrow edits were weak. Broader groups of related edits mattered more. That pushed us away from the idea of one neat hidden switch.

In practical terms:

- the broader early-market edit changed the chosen asset at a rate of **25%**, versus **16.7%** for its matched random edit
- the broader later-market edit changed the chosen asset at a rate of **33.3%**, versus **12.5%** for its matched random edit

That pointed toward a broader internal market summary rather than a single hidden dial.

With tighter controls and deterministic decoding, the result held but remained modest:

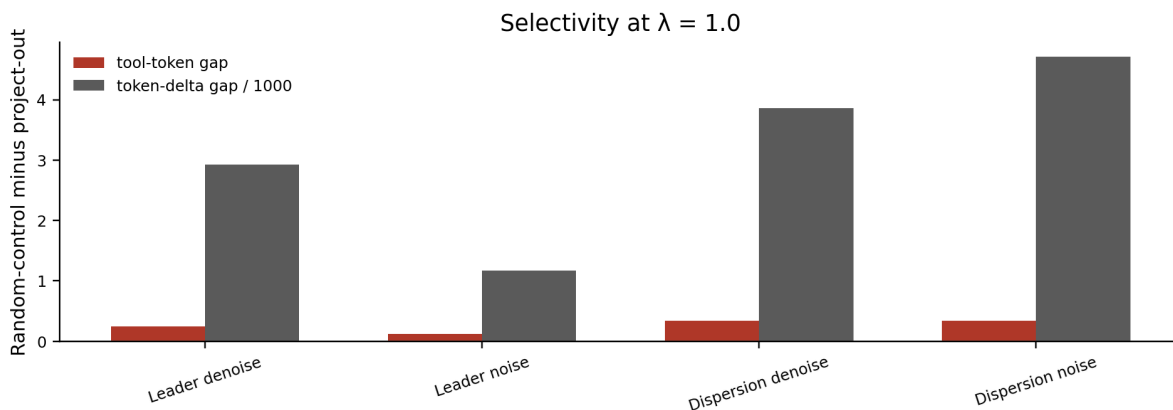
- targeted edits changed the chosen asset at **22.9%**
- matched random edits changed it at **18.8%**

Real signal, but not enough for a strong causal claim. The obvious question: are the targeted edits actually special, or would any perturbation of the same size do the same thing?

Robustness: Targeted Edits Versus Matched Random Edits

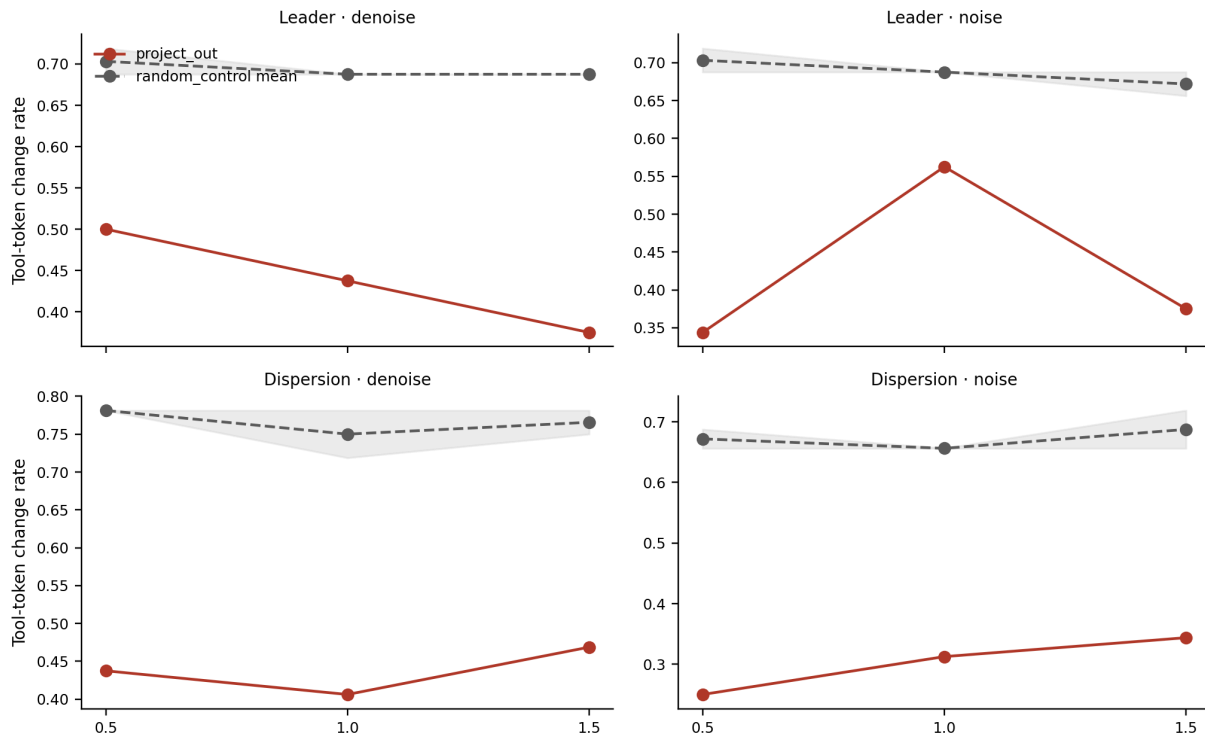
We tested this with matched prompt pairs, both weaker-side and stronger-side variants, several edit strengths, and matched random controls.

The targeted edits behaved differently from equally sized arbitrary edits.



Positive values mean the random edit was more disruptive than the targeted edit. This was the cleanest evidence that the internal market signals were real and selective rather than arbitrary noise.

In all **12 / 12** main choice comparisons, the targeted edit was less disruptive than the matched random edit. This rules out the cheapest counter-explanation. The targeted edits behaved in a structurally different way from arbitrary matched edits of the same magnitude.



Tool-token change rates across edit strengths. The targeted and random curves separate cleanly: the model's response to a targeted edit follows a different pattern than its response to a random one of the same size.

But selectivity is not causation. The signals are real and structured — but are they the reason the model makes the choices it makes?

Restoration: Can We Move Behavior Back Toward A Known Source?

The previous tests all weakened signals and watched for damage. The restoration test reversed the logic: take the internal market signal from a matched source example and transplant it into the base prompt. If the signal is truly the reason the model chose a particular asset, restoring it should push the model back toward the source example's behavior.

This is the gold standard for causal claims, and it is where most appealing stories fail.

Phase 21: Final Source-Matching Test



The stronger candidate helped a little. The weaker candidate did not. Neither behaved like the single decisive cause of the final trading choice.

The stronger candidate signal gave a modest positive result:

- agreement with the source choice rose by **+4.17 pp**
- on rows that needed help, it moved the model toward the source choice **25%** of the time
- on rows that were already right, it made things worse **6.3%** of the time

The weaker candidate signal did not hold up:

- agreement with the source choice moved by **-2.08 pp**
- its help rate was **13.6%**
- its harm rate was **15.4%**

Signal tested	Change in source agreement	Help rate	Harm rate	Verdict
Stronger market signal	+4.17 pp	25%	6.3%	Interesting, but still modest
Weaker market signal	-2.08 pp	13.6%	15.4%	Not persuasive

The restoration test is where the overall conclusion narrowed. The internal market structure is real. Context bends it. Targeted edits are selective. But none of the candidate signals we isolated could reliably move behavior back toward a known source.

Path Validation: Lesion And Rescue

One final test: instead of restoring a signal directly, damage it at an early layer and attempt a downstream rescue at a later layer using source-side coefficients. If the two sites are part of the same causal pathway, the rescue should partially undo the lesion.

The interventions applied cleanly, but the behavioral surface was too sparse to measure action-choice changes with confidence — the lesioned model did not reliably produce parsed tool calls. The path-validation question remains open for a future test at a scale where the readout is dense enough to interpret.

What We Can Now Say With Confidence

Claim	Support	Why this held up
The model builds a real internal picture of the market.	Strong	Stable market-linked internal patterns appeared in discovery and held through format decontamination, intervention tests, and robustness checks.
What the model reads before the market changes how it interprets that market.	Strong	When context moved before the market, the market-reading gap reached 0.061 for risk and 0.070 for opportunity framing. When the same context came after the market, the market read stayed effectively unchanged.
The two main internal patterns are meaningful, not random noise.	Strong	In the robustness battery, targeted edits beat matched random edits in all 12 of 12 main choice comparisons.
The stronger candidate signal is the main cause of the final trading choice.	Not supported	In the restoration test, agreement with the source choice improved by only 0.042, and the signal fixed mistakes on 0.250 of rows that needed help.
The weaker candidate signal is a convincing driver of final action choice.	Not supported	In the restoration test, agreement with the source choice fell by 0.021, while the rate at which it fixed mistakes (0.136) was slightly lower than the rate at which it created new ones (0.154).

The most important positive result is also the simplest: the model builds a real internal picture of the market, and that picture changes when earlier context shifts the frame. Everything else — the intervention tests, the robustness batteries, the restoration attempts — hangs from that central finding.

What The Evidence Does Not Support

We should not claim:

- that we found the one hidden cause of the final trading choice
- that either of the two named signals by itself explains the decision
- that every readable internal pattern is also a decisive behavioral lever

The data rules these out. The restoration test was designed specifically to catch overreach, and it did.

The Final Conclusion

FINAL CONCLUSION

The model builds a meaningful internal summary of the market, and that summary is influenced by context. We can also find recurring internal signals that line up with visible

market features. But the specific signals we isolated are not enough, on their own, to explain the model's final trading choice.

Narrower is not weaker. The data supports four things:

- the model's internal market picture is real and measurable
- prior context shapes that picture
- the model's decision is not governed by one simple hidden switch
- the two strongest candidate signals are real and selective, but not sufficient on their own to explain the final choice

These are the claims the evidence actually supports. The next round of work starts from a clear map: which questions have been answered and which remain open.